Tanuj Haresh Sidhwa

B.Tech in Mechanical Engineering (Minor: Astronomy)
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Portfolio | LinkedIn | GitHub

Research Interests: Computational Fluid Dynamics, Combustion & Propulsion, Electrohydrodynamics, Aerospace Systems, Physics-Informed Neural Networks, Reduced Order Modelling

Education

- **Indian Institute of Technology, Indore** *2023 - 2027*
  Indore, India
  - B.Tech in Mechanical Engineering, CGPA: 9.47/10
  - Minor in Astronomy, CGPA: 9.75/10

Standardized Tests

- **GRE General Test**: 323/340 (Q: 168, V: 155, AWA: 3.5)
- **TOEFL iBT**: 5.5/6 (110/120 on the legacy scale; CEFR C1)

Experience

- **Liquid Propulsion Systems Centre (LPSC), ISRO** *Jun. 2026 - Jul. 2026*
  Research Intern | Guide: Vijay Kumar, Scientist/Engineer 'SG' Bengaluru, India
  - Developed and modified an OpenFOAM-based electrohydrodynamic solver for cone-jet electrosprays, incorporating improvements to boundary conditions, interface treatment, mesh refinement, and computational efficiency.
  - Validated the computational framework against experimental cone-jet observations at LPSC, investigating Taylor-cone formation and electrospray behaviour.
  - Conducted parametric investigations of applied voltage, flow rate, electrode spacing, and emitter geometry to characterize their influence on electrospray performance.

- **IASc-INSA-NASI Summer Research Fellowship** *May 2025 - Jul. 2025*
  Research Intern | Dr. Abhijit Kushari, Advanced Propulsion Laboratory Indian Institute of Technology, Kanpur
  - Conducted static structural analysis of serpentine nozzles in ANSYS, evaluating deformation behavior under fluid-induced and inertial loading conditions relevant to propulsion and high-speed flow applications.
  - Performed fatigue, modal, and harmonic analyses to estimate component lifespan, identify resonance conditions through Campbell diagrams, and evaluate structural integrity under complex dynamic excitation environments.
  - Studied plasticity model implementation using bilinear isotropic hardening models, while developing proficiency in CATIA V5 and conducting preliminary fluid–structure interaction studies supported by literature review.

Publications & Conferences

- **1st International Conference on Thermofluids Engineering (INCOTHERM-2025)** *Oct. 2025*
  First Author IIT (ISM) Dhanbad
  - First-author paper, *"Experimental Investigations on Effects of Student Occupancy Rate in a Student Hostel Room on Indoor Air Quality,"* accepted and presented; proceedings under publication process.

- **Students' Academic Conference (SAC)** *Dec. 2025*
  Presenter Inter IIT Tech Meet 14.0, IIT Patna
  - Presented *"An Innovative Laser Decal Transfer of ZnO Ceramic in LIG for Advanced Hybrid Nanogenerator Applications"* on behalf of the primary author.

Technical Skills

- **Simulation & Aerodynamic Analysis**: ANSYS Mechanical, ANSYS Fluent, XFLR5, XFOIL, OpenFOAM
- **CAD & Design**: CATIA V5, Fusion 360, ANSYS SpaceClaim
- **Scientific Computing & Machine Learning**: Python, MATLAB, C++, NumPy, SciPy, PyTorch

Projects

- **Combustion and Exfoliation of Inorganic Gellant-Laden Kerosene Droplets** *Aug. 2026 - Present*

  B.Tech Project under supervision of Dr. Ankur Miglani
  - Studying combustion and exfoliation behavior of Silica gellant-laden kerosene droplets through controlled experiments.
  - Investigating effects of gellant composition and concentration on droplet morphology and combustion behavior.

- **Physics-Informed Neural Network for Diesel Engine** *Jan. 2026 - Present*

  Prof. Devendra Deshmukh, Department of Mechanical Engineering
  - Developed a Fourier-feature Physics-Informed Neural Network in PyTorch for predicting diesel-engine in-cylinder pressure.
  - Integrated thermodynamic constraints into the training objective to improve physical consistency, with ongoing evaluation of generalization across unseen injection profiles.

- **Computational Modeling of Propellant Sloshing Dynamics in Spacecraft Tanks** *Aug. 2026 - Present*

  B.Tech Minor Project under supervision of Dr. Bhargav Vaidya
  - Implementing an OpenFOAM-based computational model to investigate propellant sloshing in spacecraft tanks under prescribed excitation.
  - Analysing transient pressure responses and dominant sloshing frequencies to support numerical validation and reduced-order mechanical modelling.

- **Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster** *Jan. 2026 - Apr. 2026*

  Course Project: Computational Fluids and Structures (AA374), IIT Indore GitHub
  - Developed a quasi-1D finite-volume solver for compressible nozzle flows using the Euler equations.
  - Implemented Lax–Friedrichs, HLLC, and MUSCL–HLLC schemes and validated against theory and ANSYS.

- **Powered Lift Wing Design for STOL Applications** *Oct. 2025 - Dec. 2025*

  InterIIT Tech Meet 14.0: High Prep – LAT Aerospace
  - Executed CFD-driven aerodynamic design of a powered-lift STOL wing, for InterIIT Tech Meet 14.0.
  - Optimized aerodynamic performance through ANSYS Fluent studies to achieve $C_{L,\max} = 6.81$.

- **Assessment of Indoor Air Quality in Student Hostel** *Sept. 2024 - Oct. 2025*

  Prof. Ritunesh Kumar, Department of Mechanical Engineering GitHub
  - Developed an Arduino-based CO_2 monitoring system for controlled occupancy and ventilation experiments.
  - Analyzed transient CO_2 buildup trends; work resulted in a first-author paper at INCOTHERM 2025.

- **CubeSat Project** *Oct. 2024 - Nov. 2024*

  CAE Club & Astronomy Club, IIT Indore GitHub
  - Managed CubeSat structural design and cross-domain coordination within a multidisciplinary team.
  - Designed the chassis and performed structural, modal, and thermal analyses under launch loads.

Achievements

- **Reliance Foundation Undergraduate Scholar** *2024*
- **Bronze Medal, IITISoC'24 – Design & Modelling.** *2024*
- **6th Position, Inter-IIT Tech Meet 14.0 – LAT Aerospace** *2025*
- **AIR 144, GATE Aerospace Engineering** *2026*
- **Highest grade (AP), Computational Fluids and Structures (AA374), IIT Indore** *2026*

Relevant Coursework

- **Mathematics**: Linear Algebra, Differential Equations, Multivariable Calculus, Real Analysis, Complex Analysis, Numerical Methods, Calculus of Variation
- **Mechanical Engineering**: Fluid Mechanics, Thermodynamics, Heat Transfer, Solid Mechanics, Instrumentation and Control Systems, Computational Fluids and Structures, Introduction to Finite Element Methods, Fundamentals of Vibrations, Fundamentals of Acoustics
- **Astronomy Minor**: Flight Mechanics and Classical Control, High Energy Astrophysics and Transient Sky, IoT for Space Applications, Introduction to Astronomy

Positions of Responsibility

- **President**, Mechanical Engineering Student Association, IIT Indore *Apr. 2025 - Apr. 2026*
- **Team Lead**, High Prep LAT Aerospace, InterIIT Techmeet 14.0 *Oct. 2025 - Dec. 2025*
- **Project Lead**, CubeSat Project, IIT Indore *Oct. 2024 - Nov. 2024*